

Volcano

Volcanoes can look like steep cones or have gentler slopes. They might create loud, spectacular eruptions or slow and quiet lava flows. Active volcanoes erupt regularly, although predicting exactly when this will happen is difficult. There are about 1,500 potentially active volcanoes worldwide.

During an explosive eruption, ash is launched high into the air before falling back to Earth.

The ash and magma harden into layers on the sides of the volcano, building a cone shape.

One plate sinks under another.

Magma collects in underground chambers and rises, because it is less dense than solid rock.

Magma rises through chimneys that lead to openings in the Earth's crust, known as vents. A volcano can have several vents.

Coming to life

Volcanoes usually form at the edges of Earth's massive, moving tectonic plates. When two plates collide, one plate might slide under another. Magma, ash, and gases then rise up through the Earth's interior and reach the surface, creating a volcano.

Explosive eruptions

Pressure from expanding gases inside the magma chamber slowly builds up over time, before suddenly being released during an eruption. Just how explosive an eruption will be depends on the amount of gas in the magma. Magma that is sticky and full of gas typically causes violently explosive eruptions.

Volcanoes are powered by chambers deep inside the Earth that fill up with magma.

Red-hot lava In Hawaii, volcanic lava flows can reach temperatures of 2,147°F (1,175°C) and speeds of 20 mph (30 kph). At this temperature, lava glows bright red.



Underwater pillows

Hot lava that erupts from cracks on the sea floor solidifies very quickly into puffy round shapes. These lumps are known as pillow lavas.

